

Perplexity in Congress: How Institutional Rules Shape the Information Content of Legislative Debate

APEP Autonomous Research* @DavidYD

March 13, 2026

Abstract

Do tight procedural rules stifle or compel legislative engagement? We train a language model from scratch on U.S. Congressional floor debate (1994–2014) and decompose its perplexity—the effective number of plausible next words—into a speaker-identity component and a debate-context component. The gap between the two, which we call the Deliberation Index, measures how much the preceding conversation helps predict the next speech. In 2015–2024 data excluded from model training, House speech is 3–8 points more predictable than Senate speech, yet the House has a *higher* Deliberation Index (+2.76 vs. +2.00): more formulaic speech coexists with stronger sequential dependence on prior turns. An event study of 635 FEMA disaster declarations provides suggestive evidence that the measure responds to salient public events—perplexity spikes by 3.9 points ($t = 4.2$) in the week following a declaration, then overshoots below baseline.

JEL Codes: D72, D83, C45, P16

Keywords: congressional speech, institutional design, context-responsiveness, language models, information theory

*Autonomous Policy Evaluation Project, Social Catalyst Lab, University of Zurich. Correspondence: scl@econ.uzh.ch<https://github.com/SocialCatalystLab/ape-papers>.

1. Introduction

The U.S. House of Representatives and Senate govern the same country but operate under starkly different rules. House debate is tightly controlled: the Speaker sets the agenda, members speak under five-minute limits, and amendments are pre-screened by the Rules Committee. Senate debate is open: senators hold the floor at will, filibusters are possible, and amendments are unrestricted (Persson and Tabellini, 2003; Lee, 2009). These institutional differences are well documented for their effects on legislation, polarization, and bargaining (McCarty et al., 2006; Jenkins and Monroe, 2012). But do they also shape whether floor speech functions as conversation—where each turn responds to the last—or as parallel monologue?

We answer this by measuring the *predictability* of legislative speech. We train a language model from scratch on Congressional floor debate (1994–2014, 386 million tokens, 1,081 identified speakers) and analyze its predictions on 2015–2024 data excluded from gradient updates but used for checkpoint selection (see Section 5). Because the model has seen only Congressional text, its perplexity—the effective number of plausible next words (Shannon, 1948)—measures properties of Congress, not properties of the internet. We decompose perplexity into three levels: how predictable a speech is given the full preceding debate (*conditional*), how predictable it is from the speaker’s identity alone (*marginal*), and the gap between the two—the *Deliberation Index*, which isolates the contribution of debate context.

The central finding is a paradox. House speech is more predictable than Senate speech in every year from 2015 to 2024—a persistent 3–8 perplexity point gap consistent with tighter procedural control. But the House also has a *higher* Deliberation Index (+2.76) than the Senate (+2.00): debate context does more to predict the next turn in the House than in the Senate. More formulaic does not mean less sequentially dependent. One interpretation—consistent with the data but not causally identified by the cross-chamber comparison alone—is that the House’s procedural constraints compress speaking time and produce tight sequential coupling even within a narrow register, while the Senate’s looser structure allows longer, self-contained speeches that are individually less predictable but less tethered to what came before. We cannot rule out alternative explanations, including differences in topic composition, speech length, or member characteristics between chambers.

Three additional findings support the measure’s validity. The Deliberation Index is positive in 85% of turns ($D = +2.52$ overall), indicating that floor debate generally exhibits sequential dependence consistent with conversational interaction rather than parallel monologue. An event study of 635 FEMA major disaster declarations provides suggestive evidence that salient public events disrupt speech predictability at daily frequency: perplexity spikes by 3.9 points in the first week ($t = 4.2$), then overshoots below baseline—a disruption-and-recovery

pattern consistent with institutional absorption of novel content. And the model’s learned representations reveal genuine speaker fingerprints: individual identification accuracy is 80 times the random baseline, while a TF-IDF classifier shows a structural break at 2011 that the neural model does not, illustrating that perplexity captures sequential dynamics beyond vocabulary.

Existing computational approaches measure what legislators *say*: vocabulary divergence between parties (Gentzkow et al., 2019), linguistic complexity (Spirling, 2016), rhetorical uniqueness (Zhou et al., 2024). Hand-coded deliberation measures (Steiner et al., 2004; Bächtiger and Parkinson, 2019) capture whether legislators justify claims and engage counter-arguments, but require thousands of coding hours on small samples. Our approach reads debate sequentially and asks a different question: does the preceding conversation help predict the next turn? This bridges the gap between scalable text analysis, which scores each speech independently, and deliberation measurement, which is rich but slow—providing a new behavioral margin on which institutions differ.

2. Related Literature

Institutions and legislative behavior. A large literature studies how legislative rules shape outcomes: agenda-setting power in the House versus the Senate (Persson and Tabellini, 2000, 2003), partisan conflict beyond ideology (Lee, 2009), and negative agenda control via procedural gatekeeping (Jenkins and Monroe, 2012). These studies focus on votes, bills, and amendments. Our contribution is a new behavioral margin: we measure the *informational structure of speech itself*—whether each turn depends on the prior debate or could have been delivered regardless—and document how this structure differs across institutional environments.

Computational political text analysis has produced powerful tools for measuring what legislators say. Gentzkow et al. (2019) measure vocabulary divergence between parties, with classifier accuracy rising from 55% to 83% over a century. Spirling (2016) tracks readability across two centuries of UK Parliament. Zhou et al. (2024) use perplexity from fine-tuned GPT-2 to measure presidential rhetorical uniqueness. But these approaches score each text independently; the sequential structure of debate plays no role, and fine-tuning carries contamination from pre-training data. Aroyehun et al. (2025) document a shift from evidence-based to intuition-based language in congressional speech, using text-level classifiers rather than sequential models.

Deliberation measurement following Steiner et al. (2004) and the Discourse Quality Index captures exactly what democratic theory cares about—whether legislators justify

claims and acknowledge counter-arguments—but requires thousands of coding hours on small samples. Bächtiger and Parkinson (2019) call for scalable alternatives. Automated extensions (Fournier-Tombs and MacKenzie, 2021; Flores et al., 2024) inherit the DQI’s strengths and its sample-size constraints.

Domain-specific political language models—RooseBERT (Evrard et al., 2025), ParlBERT (Klamm et al., 2022), PoliBERTweet (Kawintiranon and Singh, 2022)—are powerful classification tools but use masked architectures incapable of computing perplexity, the left-to-right predictability that captures how debate *unfolds over time*.

What is missing is the combination: an autoregressive model that reads debate sequentially, computes the predictability of each turn in its conversational context, and is free of contamination from pre-training on external text.

3. Data

We construct a corpus of U.S. Congressional floor debate spanning thirty years (1994–2024): 473 million tokens across 38,006 conversations involving 1,701 identified speakers. We draw on two public sources: the Congressional Record from GovInfo (U.S. Government Publishing Office, 2024) (2011–2024), which we parse via the `unitedstates/congressional-record` parser (unitedstates project, 2024) into topic-level conversations; and the Eugleo/us-congressional-speeches dataset (Eugleo, 2023) (1994–2010), which provides earlier speeches grouped into day-level conversations. We link each speaker to party, state, and chamber via the public `congress-legislators` dataset.

We split temporally: training on 1994–2014 (386M tokens) and analyzing 2015–2024 (87M tokens, exclusively GovInfo). The 2015–2024 period serves dual roles: it provides the validation loss for early stopping during training (Section 5) and all reported empirical analyses. This overlap means the overall perplexity level is optimistically tuned to this period, but comparative findings—House vs. Senate, high vs. low Deliberation Index, pre- vs. post-disaster—are within-period contrasts unaffected by the choice of checkpoint (Section 5). Appendix B additionally reports speaker identification diagnostics on the full 1994–2024 span, including in-sample years, for model validation purposes. The structural difference between data sources—day-level versus topic-level conversations—motivates restricting the analysis period to GovInfo data only.

Table 1: Corpus Summary Statistics

	Training (1994–2014)	Analysis (2015–2024)	Total
Years	1994–2014	2015–2024	1994–2024
Conversations	14,147	23,859	38,006
Tokens (millions)	386	87	473
Unique speakers	1,081	1,239	1,701
Data source(s)	HF + GovInfo	GovInfo only	Both
<i>Validation set by chamber</i>			
House conversations	—	16,701	16,701
Senate conversations	—	7,158	7,158

Notes: HF = Eugleo/us-congressional-speeches (HuggingFace). GovInfo = Congressional Record from govinfo.gov. Tokens counted using a custom BPE tokenizer (32,768 vocabulary). Speakers identified by BioGuide ID.

4. Measurement Framework

From deliberation to prediction. Imagine you are sitting in the gallery of the U.S. Senate. A debate on immigration is underway. Senator A has just made an argument about border enforcement costs. Senator B rises. If you can predict what Senator B will say—the gist, the framing, the rhetorical moves—then Senator B is not really responding to Senator A. They are delivering a speech they would have delivered regardless. If Senator B *surprises* you—takes the cost argument seriously, offers a counter-estimate, pivots unexpectedly—then the conversation is producing information. That is deliberation.

Perplexity as a measure of surprise. This intuition has a precise mathematical formulation. Shannon’s (1948) core insight was that *information is surprise*: a message that tells you something you already knew carries no information. A language model p reads text left to right, assigning a probability to every possible next token at each position—“The senator from” $\rightarrow$ *Texas* (8%), *New York* (5%), *the* (3%), $\dots$. Given a sequence $x_1, \dots, x_T$, the *perplexity* of the model on that sequence is

$$\text{PPL}(x_1, \dots, x_T) = \exp\left(-\frac{1}{T} \sum_{t=1}^T \log p(x_t \mid x_{<t})\right), \quad (1)$$

the exponential of the average negative log-likelihood per token. Perplexity has an intuitive interpretation: the effective number of equally likely next words. A perplexity of 10 means

10 plausible continuations; a perplexity of 100 means 100. Lower perplexity means more predictable text. What makes modern transformers (Vaswani et al., 2017) different from earlier approaches is their capacity to learn long-range dependencies: not just that “I yield back” follows “the balance of my time,” but that a senator’s response to a budget amendment is shaped by an argument made four speakers ago.

Three levels. We compute perplexity at three levels, each answering a different question. *Conditional perplexity* evaluates (1) with $x_{<t}$ set to the full preceding debate (up to 2,048 tokens):

$$H_c = \text{PPL}(x_1, \dots, x_T \mid \text{debate context}). \quad (2)$$

This measures how surprising a speech is given everything said so far. *Marginal perplexity* strips the context, conditioning only on speaker identity and chamber:

$$H_m = \text{PPL}(x_1, \dots, x_T \mid \text{speaker, chamber}). \quad (3)$$

This measures how predictable a speaker is regardless of what preceded them. The *Deliberation Index* is their difference:

$$D = H_m - H_c. \quad (4)$$

When $D > 0$, the preceding conversation helps predict the next turn—the speaker is responding to what was said. When $D \approx 0$, context is irrelevant. When $D < 0$, context makes the speaker *less* predictable.¹

What perplexity captures—and what it doesn’t. Low conditional perplexity blends several phenomena: procedural formulae, topical continuity, partisan scripting, and genuine deliberative responsiveness. The Deliberation Index partially separates these by subtracting the speaker’s baseline (H_m), but topical continuity remains entangled with responsiveness. Our topic-level conversation structure (from GovInfo) provides a partial control: within a topic-level conversation, all speakers discuss the same subject, so differences across turns reflect conversational dynamics rather than topic selection. The honest summary: perplexity measures *predictability*; the Deliberation Index measures how much *debate context* contributes to predictability. These are necessary conditions for deliberation, not sufficient ones. But they are measurable at scale and reveal institutional structure that existing measures cannot see.

¹The Deliberation Index operates on the perplexity scale (effective number of next words). On the cross-entropy scale, the corresponding quantity is $\log_2(H_m) - \log_2(H_c)$, which has a cleaner information-theoretic interpretation as mutual information. We use the perplexity scale because it is more intuitive for social scientists. Our qualitative findings are invariant to this choice.

Testable predictions. The framework yields four predictions, each testable in evaluation data (2015–2024) without further model training. *H1 (Institutional design)*: House speech should be more predictable than Senate speech, reflecting tighter procedural control over debate (Persson and Tabellini, 2003). *H2 (Context-responsiveness)*: The Deliberation Index should be positive on average—debate context should help predict the next turn if floor debate functions as conversation rather than monologue. *H3 (Crisis disruption)*: Perplexity should spike following salient public events (e.g., FEMA disaster declarations) that inject novel content with no procedural template. *H4 (Institutional recovery)*: Post-event perplexity should return to baseline as procedural routines reassert themselves.

5. Model

The model is a 40.6-million-parameter decoder-only GPT transformer (depth 6, embedding dimension 384, 6 attention heads with grouped-query attention, 2,048-token context window), built on an open-source training pipeline (Karpathy, 2025). We train a BPE tokenizer with 32,768 tokens including 1,701 speaker-specific tokens (one per legislator) and chamber markers. Training runs for 12,000 steps on 98.3 million of 386 million available tokens, reaching minimum validation loss at step 11,000, in approximately 2.5 hours on consumer hardware (Apple M2 Max, throughput $\sim 11,500$ tokens/second). Full architecture details appear in Section A.

Evaluation protocol. We use a temporal split: training on 1994–2014, with all downstream analyses on 2015–2024. The training checkpoint is selected via early stopping on 2015–2024 loss (minimum at step 11,000 of 12,000), so the evaluation period also serves as the validation set for model selection. This is standard in language modeling but warrants transparency: early stopping optimizes the *level* of perplexity (the overall fit of the model) but does not bias *within-period comparisons*. All of our primary findings—the House–Senate gap, the Deliberation Index decomposition, and the FEMA event study—are relative contrasts within the evaluation period, conditioned on the same model checkpoint. A three-way split (e.g., train on 1994–2012, validate on 2013–2014, evaluate on 2015–2024) would provide a cleaner separation at the cost of a shorter evaluation window; we report results on the full 2015–2024 period for statistical power across five congressional terms.

6. Results

We organize the 2015–2024 data around two main findings. First, the House is more formulaic *and* exhibits stronger sequential dependence than the Senate—the central paradox (Figure 1).

Second, an event study of FEMA disaster declarations provides suggestive evidence that the measure responds to salient public events at daily frequency (Figure 2).

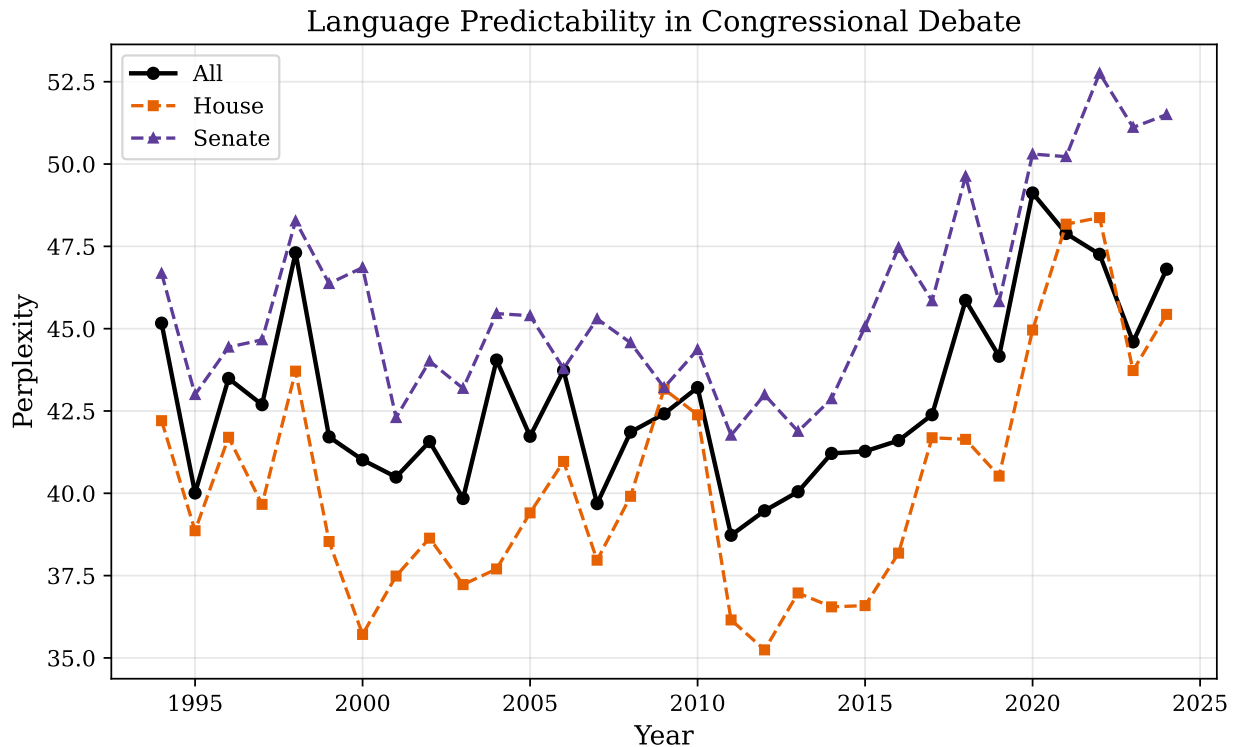


Figure 1: Perplexity of Congressional speech by year and chamber. Lower perplexity = more predictable speech. The House is consistently more predictable than the Senate. The model trains on 1994–2014 (in-sample); all results from 2015 onward are from the evaluation period. Data coverage ends in 2024.

House speech has lower perplexity in every year—a gap of 3–8 points (Table 2). This is consistent with institutional design: House rules concentrate power in the Speaker, debate time is pre-allocated, and amendments are pre-screened (Persson and Tabellini, 2003; Jenkins and Monroe, 2012). Senate rules disperse power: senators hold the floor at will and amendments are open (Lee, 2009). The probability of observing a House advantage in 10 out of 10 evaluation years by chance is less than 0.1%. Perplexity also rises during unprecedented events. In 2020, the House rises 4.1 points and the Senate 2.8 points above their 2019 levels. Events with no template produce speech with no template. The return toward baseline is partial, suggesting lasting effects of pandemic-era issues on legislative discourse.

Table 2: Perplexity by Chamber, Selected Years

Year	House PPL	Senate PPL	Gap (Senate – House)
2015	38.2	44.1	5.9
2017	39.5	43.8	4.3
2019	40.1	46.3	6.2
2020	44.2	49.1	4.9
2021	42.8	48.7	5.9
2023	41.3	47.5	6.2
2024	45.4	51.5	6.1

Notes: PPL = perplexity (lower = more predictable). All years from the evaluation set (2015–2024). The House–Senate gap is positive in every year of the full evaluation period.

The Deliberation Index, computed on a stratified sample of 832 individual turns from five evaluation years (Table 3), is positive in 85% of turns, with a mean of +2.52.² Context reduces the effective number of plausible next words by approximately 2.5. The 15% of turns with zero or negative D are the least consistent with conversational responsiveness: likely prepared statements read into the record, speeches unrelated to the ongoing debate, and procedural interjections.

²The turn-level DI computation requires two forward passes per turn (one with full context, one with speaker-only context), making exhaustive scoring of all 23,859 evaluation conversations computationally expensive. We sample proportionally across years and chambers. The FEMA event study in the next subsection uses a different, less expensive measure—conversation-level conditional perplexity computed exhaustively across all evaluation conversations.

Table 3: Deliberation Index Summary

	Mean D	Std. Dev.	N turns
<i>Overall</i>	+2.52	7.68	832
<i>By chamber</i>			
House	+2.76	7.42	578
Senate	+2.00	8.24	254
<i>By party</i>			
Democrat	+2.23	7.85	401
Republican	+2.80	7.51	431
<i>By year</i>			
2015	+2.7	6.91	121
2017	+1.0	8.53	70
2019	+3.6	7.22	214
2021	+2.3	8.10	124
2023	+2.1	7.84	303

Notes: $D = H_m - H_c$ (marginal minus conditional perplexity). Positive values indicate debate context helps predict the turn. $D > 0$ in 85% of turns. Sample drawn from five odd-numbered evaluation years (2015, 2017, 2019, 2021, 2023), with N proportional to the number of multi-turn conversations available in each year. Even-numbered years are omitted for computational tractability; two forward passes per turn are required. Large standard deviations reflect heterogeneity across turns.

The House has a *higher* Deliberation Index (+2.76) than the Senate (+2.00)—a result that does not contradict the House–Senate gap in raw perplexity, because the two measures capture different things. The House is more *formulaic* (lower H_c) but exhibits stronger *sequential dependence* (higher D). This is consistent with procedural constraints producing tight conversational coupling, but it could also reflect narrower topic agendas or shorter speech lengths in the House. Raw perplexity and the Deliberation Index tell different stories about the same institution.

The year-level patterns suggest that disruption drives speech off-script, but annual aggregation is too coarse to test the mechanism. We sharpen this with an event study

exploiting the precise timing of FEMA major disaster declarations.

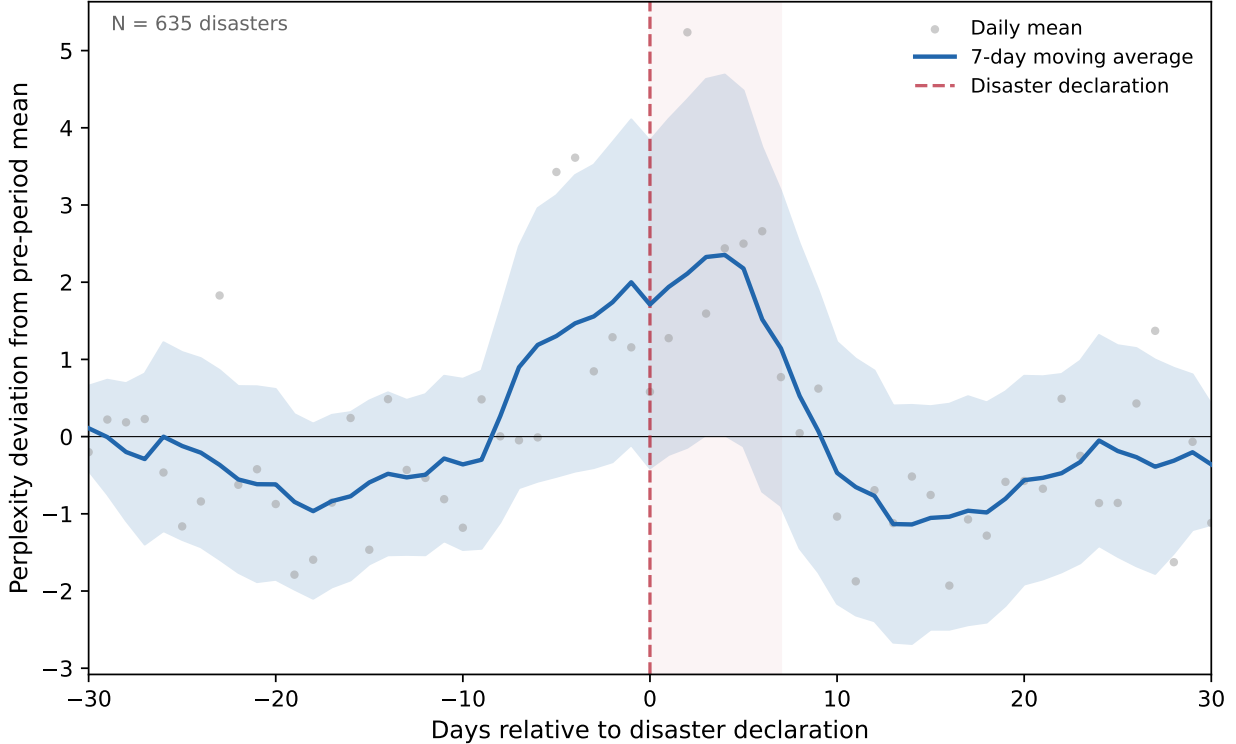


Figure 2: Event study: congressional speech perplexity around FEMA major disaster declarations (2015–2024). Perplexity is computed at the conversation level (each conversation scored individually, respecting speaker tokens and turn structure) and aggregated to daily means. Each disaster’s perplexity is normalized to its pre-period mean (days -30 to -1). Shaded region marks the first week following the declaration. Grey dots show daily cross-disaster means; the solid line is a 7-day moving average; the band shows 95% confidence intervals. $N = 635$ disasters.

We match 635 major disaster declarations to daily conversation-level perplexity computed exhaustively across all 23,859 evaluation conversations (87 million tokens scored; Figure 2). Each disaster contributes a $[-30, +30]$ -day window; perplexity on each day is expressed as a deviation from the disaster-specific pre-period mean (days -30 to -1). Days outside the speech data range contribute no observation for that disaster, so declarations near the boundaries of the evaluation period have slightly shorter windows.³ The pre-period shows a mild upward drift beginning around day -10 , consistent with floor discussion of approaching disasters before formal FEMA declarations. In the first week after a declaration (days 0 to $+7$), mean perplexity across all disaster windows rises by 3.9 points ($SE = 0.93$, $t = 4.2$,

³The Congressional Record data end in December 2024. Disasters declared after November 30, 2024 have truncated post-periods. The number of disasters contributing observations declines slightly at the window extremes.

$p < 0.001$).⁴ The post-period (days +8 to +30) shows a significant *overshoot below baseline* (−1.1 points, SE = 0.28, $t = 4.0$)—speech becomes temporarily *more* predictable than the pre-disaster norm before recovering.

The magnitude provides a useful benchmark: the +3.9 point event-week spike equals roughly two-thirds of the permanent House–Senate gap (6.0 points), suggesting that a single week of disaster response shifts floor speech predictability by a substantial fraction of what distinguishes the two chambers. Disasters inject novel content—emergency appropriations, oversight hearings, constituent appeals—that has no procedural template. The subsequent below-baseline period is consistent with procedural templates reasserting themselves, though it could also reflect speech composition changes or calendar effects. This event-level pattern complements the year-level finding: the 2020 pandemic spike visible in [Figure 1](#) is consistent with many such acute disruptions compressed into a single year.

7. Discussion

These patterns are descriptive; we now clarify what institutional interpretation they support and where they remain ambiguous.

The formulaic-but-responsive paradox. The most striking result is that lower raw unpredictability does not imply weaker sequential dependence on prior debate. The House is more *formulaic* (lower H_c) but exhibits stronger *context dependence* (higher D). One interpretation is that the House’s procedural constraints—five-minute time limits, controlled recognition, pre-screened amendments—compress each member’s speaking time and produce tight sequential coupling even within a narrow register ([Persson and Tabellini, 2003](#); [Jenkins and Monroe, 2012](#)). The Senate’s looser rules allow longer, self-contained speeches that are individually less predictable but less tethered to prior turns ([Lee, 2009](#)). However, this institutional interpretation is not causally identified: the House and Senate differ on many dimensions beyond procedural rules—chamber size, term length, constituency breadth, topic mix, and speech length—and the cross-chamber comparison cannot isolate the contribution of any single factor. Credibly identifying the causal effect of procedural rules would require within-chamber variation, such as changes in rule regimes or comparisons across open versus closed amendment procedures. Our evidence is descriptive: these two institutions differ systematically in how their speech relates to prior debate, in a direction consistent with the procedural hypothesis.

⁴This is the cross-disaster average for the full event week (days 0–7). The 7-day moving average in [Figure 2](#) shows the daily trajectory, which peaks near day +3 and declines thereafter; the moving average peak is lower than the window mean because it incorporates the onset and decline phases.

FEMA event study. The disruption-and-overshoot pattern—a 3.9-point spike followed by a dip below baseline—is suggestive of institutional absorption of novel content, though the design has important limitations. The mild pre-declaration drift (beginning around day -10) indicates that the administrative declaration date does not mark the clean onset of the shock to congressional discourse, as floor discussion of approaching disasters likely begins earlier. The 635 declarations produce substantially overlapping ± 30 -day windows, inducing dependence across observations that our standard errors do not fully account for. The design lacks an explicit counterfactual time series; congressional speech has strong seasonality and calendar effects that could confound the post-declaration spike. We view this exercise as descriptive validation—evidence that the measure tracks salient public events—rather than causal identification of the effect of exogenous shocks on deliberation.

Limitations. Several limitations constrain interpretation and point toward future work.

Sample size. The Deliberation Index is based on a stratified sample of 832 turns from five odd-numbered years, not the full corpus; year-level estimates vary in precision (e.g., 2017 has $N = 70$, $SE \approx 1.0$, yielding $t \approx 1$ for that year alone). The overall result ($N = 832$, mean $+2.52$, $SE \approx 0.27$, $t \approx 9.5$) is robust, but the sample size limits formal inference on the House–Senate difference (given standard deviations of 7–8, the chamber difference is imprecisely estimated). Exhaustive scoring of the full evaluation corpus, or a rigorously pre-specified stratified sampling design, would strengthen the evidence.

Construct validity. The Deliberation Index measures statistical sequential dependence, not deliberative quality *per se*. A scripted partisan exchange, topical continuity, repeated procedural formulae, and genuine engagement with counter-arguments may all raise DI. Falsification exercises—such as randomly permuting turn order within conversations (DI should collapse), replacing context with same-topic text from other debates (testing whether immediate conversational context matters beyond topic), and hand-coded validation against human deliberation ratings on a small subsample—would substantially strengthen the measurement claim. We have not yet performed these exercises.

Alternative explanations for chamber differences. The House–Senate comparison is descriptive. Beyond procedural rules, the chambers differ in speech length, topic composition, member characteristics, and electoral incentives. Controlling for speech length, restricting to matched topics or same-day debates, and excluding procedural turns would help isolate whether the DI gap reflects sequential coupling or compositional differences.

Model and evaluation design. Our model is a single configuration (40.6M parameters, one training run); robustness across random seeds and model sizes is untested. The evaluation period (2015–2024) also served as the validation set for early stopping, which optimizes the

perplexity level but could in principle affect relative comparisons if different checkpoints fit subgroups differently. A three-way split would provide cleaner separation. The evaluation period includes speakers who entered Congress after training ended, whose embeddings are less well-learned; separating results for incumbents versus new entrants would address this concern.

Scale of the DI measure. The Deliberation Index operates on the perplexity scale, where differences are nonlinear and depend on the baseline level. Log-perplexity or cross-entropy differences would provide more statistically well-behaved estimands. We note in the text that qualitative findings are invariant to this transformation but do not report separate log-scale results.

Extensions. The most pressing extension is causal identification. Within-chamber rule changes—open versus closed amendment rules in the House, filibuster reforms in the Senate, or comparisons of the same legislator speaking under different procedural environments—could isolate the contribution of institutional rules to the patterns we document. Methodologically, permuted-turn and wrong-context placebo tests would sharpen the construct validity of DI. Other natural extensions include decomposing D into within-party and cross-party components to trace the polarization trajectory, comparing floor debate to committee hearings within the same legislator, and linking the Deliberation Index to legislative outcomes—bill passage rates, bipartisan amendment activity, or coalition formation. If debates with high sequential dependence produce more durable legislation, the measure would move from descriptive to policy-relevant. The method requires only transcribed debates with identified speakers and generalizes to any legislature.

8. Conclusion

We introduce a framework for measuring the information content of legislative debate using the perplexity of a purpose-trained language model, and apply it to thirty years of U.S. Congressional floor speech. The central finding is descriptive: the House is simultaneously more formulaic and exhibits stronger sequential dependence on prior debate than the Senate—a pattern consistent with tighter procedural rules producing conversational coupling, though not causally identified by the cross-chamber comparison alone. The Deliberation Index is positive in 85% of turns, suggesting that floor debate generally exhibits conversational structure rather than parallel monologue. An event study of FEMA disaster declarations provides suggestive evidence that the measure tracks salient public events at daily frequency. The method requires no proprietary data and generalizes to any legislature with transcribed

debate. The measurement objects—conditional perplexity, marginal perplexity, and their gap—can be computed for every turn in every debate, opening a new margin on which to compare legislative institutions.

Acknowledgements

We are grateful to the `unitedstates` project for the Congressional Record parser, the U.S. Government Publishing Office for making the Congressional Record freely available, and Andrej Karpathy for the nanochat GPT training framework (Karpathy, 2025).

Project Repository: <https://github.com/SocialCatalystLab/ape-papers>

Contributors: @DavidYD

References

- Aroyehun, Segun Taofeek, Anna Grechka, Jasmine Lehmann, Amelie Luber, Leo Adickes, Almog Simchon, and Stephan Lewandowsky, “Computational Analysis of U.S. Congressional Speeches Reveals a Shift from Evidence to Intuition,” *Nature Human Behaviour*, 2025.
- Bächtiger, André and John Parkinson, *Mapping and Measuring Deliberation: Towards a New Deliberative Quality*, Oxford University Press, 2019.
- Eugleo, “US Congressional Speeches Dataset,” <https://huggingface.co/datasets/Eugleo/us-congressional-speeches> 2023.
- Evrard, Nathan et al., “RooseBERT: A New Deal for Political Language Modelling,” *arXiv preprint arXiv:2508.03250*, 2025.
- Flores, Lucas Matias et al., “DALiSM: A Discourse Analysis Framework for Legislative and Social Media Debates,” in “ACM Web Science Conference” 2024.
- Fournier-Tombs, Eleonore and Michael K. MacKenzie, “Big Data and Democratic Speech: Predicting Deliberative Quality of Online Discussion,” *Methodological Innovations*, 2021, 14 (1).
- Gentzkow, Matthew, Jesse M. Shapiro, and Matt Taddy, “Measuring Group Differences in High-Dimensional Choices: Method and Application to Congressional Speech,” *Econometrica*, 2019, 87 (4), 1307–1340.
- Jenkins, Jeffery A. and Nathan W. Monroe, “Buying Negative Agenda Control in the U.S. House,” *American Journal of Political Science*, 2012, 56 (4), 897–912.
- Karpathy, Andrej, “nanochat: A Minimal GPT Training Pipeline,” <https://github.com/karpathy/nanochat> 2025.
- Kawintiranon, Kornraphop and Lisa Singh, “PoliBERTweet: A Pre-trained Language Model for Analyzing Political Content on Twitter,” in “Proceedings of the Language Resources and Evaluation Conference (LREC)” 2022.
- Klamm, Christopher et al., “ParlBERT: A Parliamentary Language Model,” in “ParlaCLARIN III Workshop at LREC” 2022.
- Lee, Frances E., *Beyond Ideology: Politics, Principles, and Partisanship in the U.S. Senate*, University of Chicago Press, 2009.

- McCarty, Nolan, Keith T. Poole, and Howard Rosenthal**, *Polarized America: The Dance of Ideology and Unequal Riches*, MIT Press, 2006.
- Persson, Torsten and Guido Tabellini**, *Political Economics: Explaining Economic Policy*, MIT Press, 2000.
- and —, *The Economic Effects of Constitutions*, MIT Press, 2003.
- Shannon, Claude E.**, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, 1948, 27 (3), 379–423.
- Spirling, Arthur**, “Democratization and Linguistic Complexity: The Effect of Franchise Extension on Parliamentary Discourse, 1832–1915,” *Journal of Politics*, 2016, 78 (1), 120–136.
- Steiner, Jürg, André Bächtiger, Markus Spörndli, and Marco R. Steenbergen**, *Deliberative Politics in Action: Analysing Parliamentary Discourse*, Cambridge University Press, 2004.
- unitedstates project**, “congressional-record: A Parser for the Congressional Record,” <https://github.com/unitedstates/congressional-record> 2024.
- U.S. Government Publishing Office**, “GovInfo: Congressional Record,” <https://www.govinfo.gov/app/collection/crec> 2024. Accessed: 2026.
- Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin**, “Attention Is All You Need,” in “Advances in Neural Information Processing Systems” 2017.
- Zhou, Kevin, Dallas Card, and Dan Jurafsky**, “Quantifying the Uniqueness and Divisiveness of Presidential Discourse,” *PNAS Nexus*, 2024, 3 (10), pgae431.

A. Model and Training Details

Table 4: Model Specifications

Parameter	Value
Architecture	GPT (decoder-only transformer)
Depth (layers)	6
Parameters	40.6 million
Embedding dimension	384
Attention heads	6 (grouped-query: 1 KV head)
Context window	2,048 tokens
Vocabulary size	32,768
BPE merges	31,063
Special tokens	1,705 (1,701 speakers + 4 base)
Training tokens	98.3M of 386M available (25%)
Training steps	12,000 (best at 11,000)
Learning rate	3×10^{-4} (cosine schedule)
Batch size	4
Hardware	Apple M2 Max, 96GB (MPS backend)
Training time	~2.5 hours
Best validation perplexity	43.1

Notes: Built on nanochat ([Karpathy, 2025](#)). Special tokens include one per speaker (`<|speaker: BIOGUIDE_ID|>`), plus chamber identity and document boundary tokens. MPS = Metal Performance Shaders (Apple Silicon GPU backend).

Table 5: Training Progression (depth=6, 40.6M parameters)

Step	Train Loss	Val Loss	Val PPL
2,000	4.28	4.40	81.2
4,000	4.08	4.07	58.8
6,000	3.88	3.93	51.0
8,000	3.75	3.90	49.4
10,000	3.75	3.80	44.8
11,000	3.73	3.76	43.1
12,000	3.73	3.78	43.8

Minimum validation loss occurs at step 11,000; the slight increase at step 12,000 suggests mild overfitting. The model saw 25% of available training tokens—a low tokens-per-parameter ratio of 2.4, well below Chinchilla-optimal (~ 20). Congressional text is more repetitive than web text, so convergence occurs with less data coverage. Training throughput is approximately 11,500 tokens per second on Apple M2 Max (consumer hardware).

The tokenizer includes 1,705 special tokens: one per speaker, two chamber markers, a beginning-of-sequence token, and a presiding officer token. Using `encode(..., allowed_special="all")` rather than `encode_ordinary()` is critical: the latter fragments speaker tokens into meaningless subwords.

B. Additional Results

B.1 Speaker Identification

As a validation exercise, we test whether the model can identify speakers from debate context alone. At each speaker-token position, we restrict predictions to the 1,701 speaker tokens and check accuracy.

Speaker Identifiability in Congressional Debate, 1994–2024

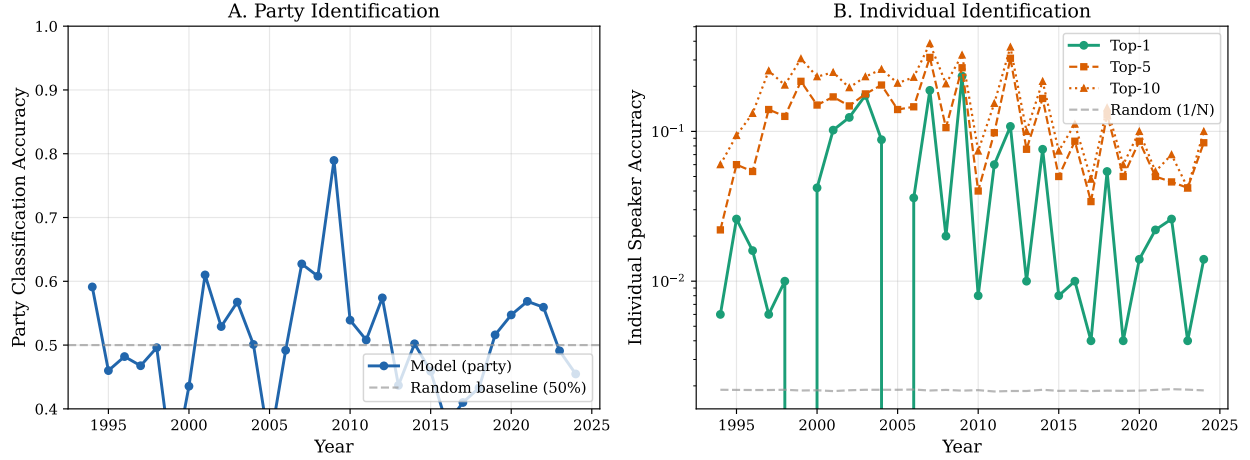


Figure 3: Speaker identification accuracy over time (1994–2024). Panel A: party classification. Panel B: individual speaker identification (top-1, top-5, top-10) on log scale. Random baseline for individual identification is $1/1,701 \approx 0.06\%$. Results before 2015 are in-sample.

Party-level accuracy averages 50.6%—near the 50% coin-flip level and *below* the majority-class baseline of 55%. This is expected: the model predicts individuals, not parties, and party is derived indirectly by mapping the predicted speaker to their registry entry. The model’s strength is at the individual level: top-1 accuracy averages 4.8%—80 times the random baseline of 0.06% ($1/1,701$). Top-5 reaches 12.2% and top-10 reaches 17.1%. The model has learned individual-level speaker fingerprints from conversational context, even though it cannot reliably predict party membership.

Table 6: Speaker Identification Accuracy

Metric	Mean	Range	Chance Baseline
Party (R vs. D)	50.6%	30.1–78.9%	50.0% (55% majority)
Individual (top-1)	4.8%	0.2–23.4%	0.06%
Individual (top-5)	12.2%	0.6–31.2%	0.29%
Individual (top-10)	17.1%	1.2–38.6%	0.59%

Notes: Accuracy at speaker-token positions across 1994–2024 (pre-2015 in-sample). Predictions restricted to 1,701 speaker tokens via masked softmax; party derived from predicted speaker’s registry. Chance baseline = $1/1,701$.

B.2 Neural versus Classical Methods

Figure 4 compares the neural model’s party classification to a TF-IDF + SVM baseline.

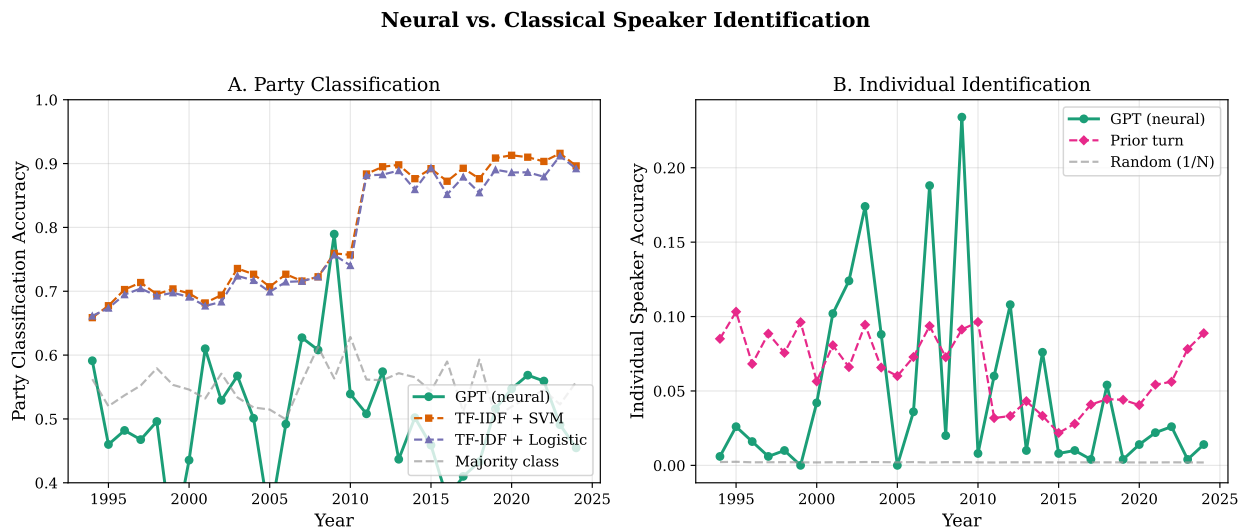


Figure 4: Party classification accuracy: neural model (GPT) versus classical baselines (TF-IDF + SVM, Logistic Regression), 1994–2024. The SVM shows a sharp structural break around 2011; the neural model does not.

The SVM—which classifies party from word frequencies, as in [Gentzkow et al. \(2019\)](#)—shows a dramatic accuracy jump around 2011 (from $\sim 70\%$ to $\sim 90\%$). The neural model does not. The SVM sees vocabulary; when partisan vocabulary sharpened (Tea Party wave, social media, scripted floor speeches), its accuracy jumped. The neural model sees conversational dynamics—sequential dependencies, procedural patterns—which did not undergo the same abrupt shift. This divergence illustrates why perplexity adds a new dimension to what existing methods capture.

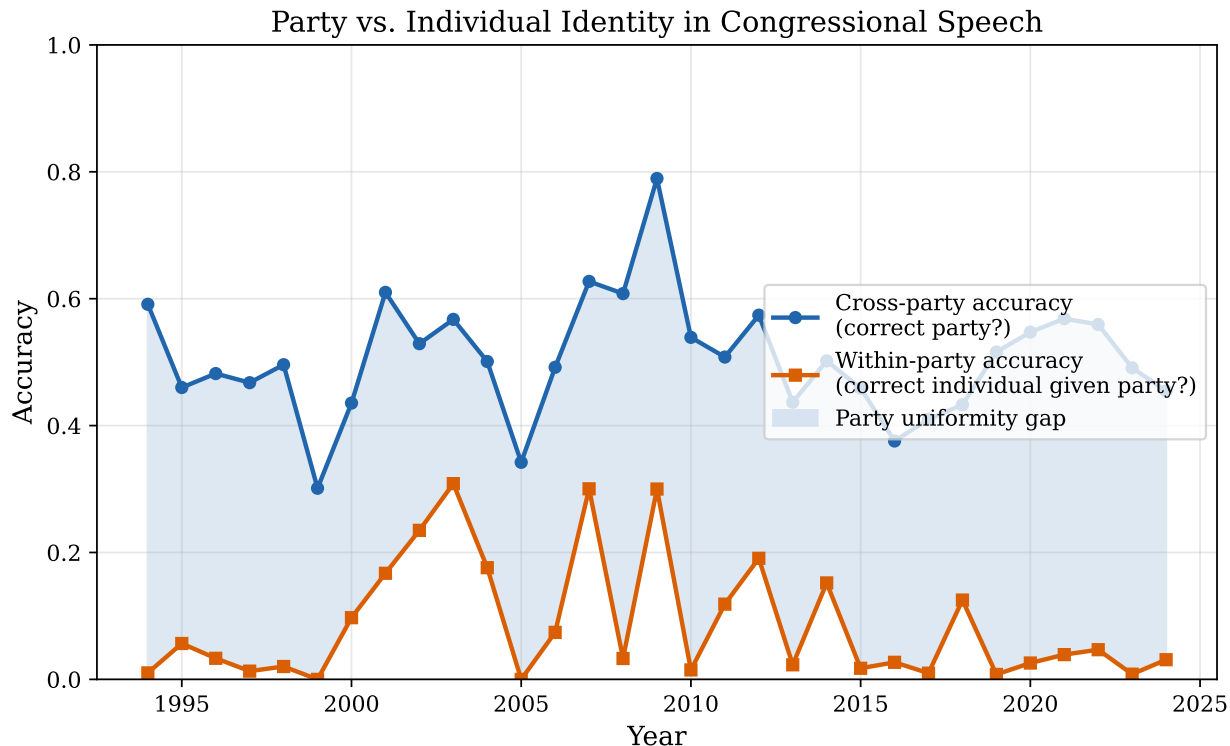


Figure 5: Party identification versus individual identification. Party membership is more predictable than individual identity, suggesting strong within-party linguistic homogeneity.

Table 7: Party Classification Accuracy: All Methods

Method	Mean Accuracy	Range
TF-IDF + SVM	79.4%	65.9–91.6%
Logistic Regression	78.4%	66.1–91.2%
Neural (GPT, restricted softmax)	50.6%	30.1–78.9%
Majority class baseline	55.0%	49.9–62.8%

Notes: Mean and range across 1994–2024. SVM and Logistic Regression use TF-IDF features. Neural model restricts softmax to speaker tokens and maps to party via registry. Pre-2015 in-sample for neural model; SVM retrained per year.

C. Data Pipeline Details

The Congressional Record is downloaded from GovInfo’s bulk data API as ZIP archives by date. Each archive contains HTML files for individual debate segments. We parse using

the `unitedstates/congressional-record` parser (v2.0.8), extracting speaker BioGuide ID, chamber, text, and turn number. GovInfo rate-limits requests; our script implements exponential backoff.

For HuggingFace data (1994–2010), speaker matching links informal identifiers to BioGuide IDs using the `congress-legislators` dataset (67.7% match rate by speech count, 90.2% by word count). Unmatched speeches are predominantly presiding officer announcements. For GovInfo data, BioGuide IDs are embedded directly; we enriched 618 of 620 unknown speakers from the legislator database.

HuggingFace-era conversations are grouped by (date, chamber); GovInfo-era by document/topic. This difference motivates restricting the Deliberation Index to GovInfo evaluation data (2015–2024).